HAPS — Wren-style hybrid stratospheric platform

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Forge project pack

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Illustrative only — not a technical specification. All renders in this document are generated for visual reference; component arrangement, proportions, and identities may differ from the final engineered assembly.

TOTALS AT A GLANCE

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UNIT COST	£1,473,750
CEILING	£2,500,000
HEADROOM	£1,026,250

Contents

1. Brief

FOUNDER SUBJECT

A European-sovereign high-altitude pseudo-satellite for persistent stratospheric operations, modelled on the Wren Aerospace Wren Flyer. Hybrid solar plus hydrogen propulsion: solar arrays sized for daytime cruise, fuel-cell-grade hydrogen storage for night and cloud-cover endurance, a regenerative buffer battery, and a space-grade composite airframe. Target operating ceiling 20 kilometres, multi-week endurance, 35-metre wingspan, 25-kilogram sensor payload (electro-optical, signals intelligence, secure 5G relay). Fixed-wing unmanned, autonomous flight controller, 1090 ES transponder, ground station with satellite uplink. Must launch from a 200-metre runway and recover via gentle stratospheric descent. Target system unit cost under 2.5 million pounds. Compliant with Eurocontrol stratospheric airspace coordination.

MISSION

To provide European-sovereign, persistent stratospheric surveillance and communications relay from 20 km altitude, operating for multiple weeks without recovery, as a recoverable and cost-effective alternative to low-Earth-orbit satellites.

USE CASE

Launched from a 200-metre runway, the platform climbs to 20 km and loiters for multiple weeks over a theatre of interest, delivering continuous electro-optical imaging, signals intelligence collection, and secure 5G relay to ground and maritime forces before recovering via powered glide.

TARGET CUSTOMERS

European defence and intelligence agencies, NATO member-state governments, critical national infrastructure operators requiring secure 5G relay, border and maritime surveillance authorities

WHY NOW

European strategic autonomy policy is driving demand for sovereign, non-US-dependent persistent overhead capability; LEO satellite congestion and launch costs make recoverable HAPS economically compelling; EASA and Eurocontrol are actively developing stratospheric airspace coordination frameworks that create a near-term regulatory pathway; hybrid solar-hydrogen propulsion has reached sufficient technology readiness for multi-week endurance at 20 km.

Constraints declared

Unit cost ceiling	£2,500,000
Max mass	—
Target process	not declared
Target material	not declared
Tolerance target	not declared
Quantity target	not declared

2. Regulatory posture

EASA CS-UAS not-started

Certified Unmanned Aircraft Systems airworthiness

Certified category standard expected at SAIL V–VI for BVLOS stratospheric operations beyond visual line of sight.

Eurocontrol SERA not-started

Stratospheric airspace coordination

Eurocontrol coordination framework governing unmanned operations in stratospheric airspace above FL600.

DO-178C not-started

Software considerations in airborne systems

Software assurance standard applied at DAL-B or DAL-C commensurate with risk; governs flight controller and mission management software.

DO-160G not-started

Environmental conditions and test procedures for airborne equipment

Environmental qualification standard covering high-altitude thermal, vibration, and EMI categories applicable to stratospheric operation down to approximately 70 °C.

ECSS-E-ST-32C not-started

Structural general requirements (space-grade composites)

Invoked by the space-grade composite descriptor; requires A/B-basis material allowables and structural test model or proto-flight model verification approach.

AS9100D not-started

Quality management systems — aerospace

Aerospace quality management standard required for manufacture of flight-critical composite and propulsion structures.

MMPDS not-started

Metallic materials properties development and standardisation

Governs allowables for metallic fittings and hard-points within the composite airframe.

3.1 · Port Wing Assembly

Module id: port_wing

28.00 kg · 20 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Left-hand high-aspect-ratio composite lifting surface carrying half the solar array, control surfaces and outboard motor mount.

WHY IT MATTERS

The wing simultaneously provides ~50% of the aircraft's lift, ~50% of its primary energy generation, and the structural attachment for the outboard propulsion unit. Any structural, electrical or aero-elastic failure of this assembly is directly mission-ending and likely airframe-loss; its stiffness and mass distribution also dominate flutter margin and CG control on a 35 m platform.

DESCRIPTION

The Port Wing Assembly is a 17.5 m half-span ultra-high-aspect-ratio composite lifting surface built around a single primary CFRP box spar with unidirectional carbon fibre caps and a Nomex honeycomb shear web, sized for the low dynamic pressure cruise condition at 20 km altitude where wing root bending moment dominates. The aerodynamic shell is a thin sandwich of T800/cyanate-ester prepreg face sheets over Rohacell foam core, vacuum-bagged in a female autoclave mould with a UV-stable outer ply to survive stratospheric solar exposure and -70 °C to +60 °C thermal cycling. The upper surface is laminated with thin-film GaAs solar cells (~12 m² active area on the port half) wired in series-parallel strings routed through the spar to a leading-edge harness exiting at the wing root.

KEY PARTS (7)

- Unidirectional CFRP spar caps (e.g. Toray M55J/cyanate-ester prepreg) with Nomex honeycomb shear web, ~17.5 m half-span
- CFRP/epoxy skin sandwich (T800 fibre, 0.3-0.5 mm face sheets over 5 mm Rohacell 51IG-F core), autoclave cured
- Thin-film GaAs solar cell laminate (e.g. MicroLink Devices IMM-±, ~32% efficiency, ~250 g/m²) bonded to upper skin, ~12 m² port half
- Outboard composite motor nacelle/pylon with titanium spar interface fittings (Ti-6Al-4V, AMS 4928)
- Aileron + outboard flaperon with Volz DA-22 high-altitude qualified servos (-70 °C rated, brushless)
- Pitot-static and AoA vane assembly (heated, stratospheric-grade, e.g. Aerosonde-class)
- Wing root joint: 4x titanium tension/shear pins with self-aligning spherical bushings to fuselage centre spar

FAILURE MODES (4)

- Wing root spar delamination or fatigue crack from repeated thermal cycling and gust loads, leading to in-flight structural failure
- Aero-elastic flutter or divergence at altitude due to insufficient torsional stiffness of the thin sandwich skin
- Solar cell laminate cracking/debonding from CTE mismatch with CFRP skin, degrading power generation
- Control surface servo seizure at -70 °C causing loss of roll authority

OPEN QUESTIONS (3)

- Final aspect ratio and chord distribution (washout/twist) pending aero-structural optimisation
- Spar cap layup thickness and ply count to meet ECSS B-basis allowables at 20 km gust case
- Solar cell string topology and MPPT partitioning between port and starboard halves

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Unidirectional CFRP spar caps (M55J/cyanate-ester prepreg) with Nomex honeycomb shear web — Custom autoclaved composite structure, 17.5m span, high-modulus fibers.	make	Autoclave Curing · Toray M55J/cyanate-ester prepreg, Nomex honeycomb	£18,500
CFRP/epoxy skin sandwich (T800 fibre, Rohacell 51IG-F core) — Autoclave-cured sandwich skins, large area, high-quality aerospace process.	make	Autoclave Curing · T800 carbon fibre/epoxy, Rohacell 51IG-F foam	£8,500
Thin-film GaAs solar cell laminate (MicroLink Devices IMM-±, ~12m²) — High-efficiency space-grade solar cells, ~£10000/m² for 12m².	buy	—	£120,000
Outboard composite motor nacelle/pylon with titanium spar interface fittings (Ti-6Al-4V) — Custom composite nacelle with machined titanium fittings.	buy	—	£3,200
Aileron + outboard flaperon with Volz DA-22 high-altitude servos (×2) — Custom composite control surfaces with purchased servos.	buy	—	£5,800
Volz DA-22 high-altitude servos (×2, purchased for integration) — Stratospheric-rated brushless servo, ~£3000 each.	buy	—	£6,000
Pitot-static and AoA vane assembly (heated, stratospheric-grade) — Specialised aerospace probe assembly, heated type.	buy	—	£1,500
Wing root joint: 4x titanium tension/shear pins with self-aligning spherical bushings — High-strength aerospace fasteners and bushings, precision components.	buy	—	£2,400

Labour £12,000 — Skilled composite fab & assembly, ~300 hrs at £40/hr.

Per-unit total **£177,900**

ASSUMPTIONS

- Solar cell cost based on space-grade efficiency.
- Autoclave cycles are major cost driver.
- Servo pricing from high-altitude UAV suppliers.

High-cost advanced composite wing with space-grade solar cells.

3.2 · Starboard Wing Assembly

Module id: starboard_wing 28.00 kg · 20 wk lead · Specialist judgement · mirrors port_wing

No render generated yet for this module.

PURPOSE

Right-hand mirrored composite lifting surface carrying half the solar array, control surfaces and outboard motor mount.

WHY IT MATTERS

At 35 m span and 20 km cruise altitude, wing structural efficiency directly governs MTOM, endurance and altitude capability. The wing is simultaneously the primary lifting surface, half of the solar power generator, and the outboard propulsion mount — any mass growth or stiffness shortfall cascades into reduced payload, lower ceiling, or aeroelastic instability (flutter/divergence) that can destroy the aircraft.

DESCRIPTION

The starboard wing is a 17.5 m half-span ultra-high-aspect-ratio cantilever composite structure mirroring the port wing. It comprises a single primary CFRP I-beam spar (unidirectional M55J-class fibre caps with ±45° shear web), foam-core ribs at ~400 mm pitch, and thin sandwich skins (CFRP face sheets over Nomex/Rohacell core) forming a torsion box. The upper skin is laminated with high-efficiency IBC silicon photovoltaic cells under ETFE encapsulation, generating the starboard half of the solar DC bus. An outboard nacelle carries the wing-mounted brushless motor and propeller, with loads reacted through titanium hard-points bonded into the spar.

KEY PARTS (8)

- Unidirectional CFRP spar caps (e.g. Toray M55J/cyanate-ester prepreg) forming wing root-to-tip primary spar, ~17.5 m half-span
- CFRP/Nomex honeycomb sandwich skin panels, sub-millimetre face sheets, autoclave-cured in female mould (UV-stable outer ply)
- Wing-laminated thin-film/IBC mono-Si solar cell array (~25-28% efficiency, e.g. SunPower Maxeon Gen III or AzurSpace flexible cells), bonded to upper skin with ETFE encapsulation
- Outboard composite motor nacelle with titanium hard-points bonded to spar, sized for ~1-2 kW brushless motor
- Twin CFRP control surface set: aileron and outboard spoileron, hinged on Kevlar-reinforced flexures
- Volz DA-22 or equivalent stratospheric-rated brushless servo actuators (qualified to -70 °C, sealed)
- Pre-preg foam-core ribs (Rohacell 71IG-F) at ~400 mm pitch with ply drop-offs at root
- Wing root titanium fitting (Ti-6Al-4V) with shear pins for fuselage carry-through interface

FAILURE MODES (4)

- Aeroelastic flutter or static divergence due to insufficient torsional stiffness at low stratospheric dynamic pressure
- Solar cell delamination or micro-cracking from thermal cycling (-70 °C to +60 °C) and UV exposure over multi-week missions
- Spar cap fibre compression failure at root under gust loading during low-altitude climb-out
- Servo or wiring harness failure at -70 °C causing control surface runaway or lock

OPEN QUESTIONS (3)

- Final aspect ratio and chord distribution (driving spar sizing and rib spacing)
- Whether outboard motor mount is required on this wing or only on port (single vs twin propulsion configuration not published)
- Solar cell technology selection — rigid IBC vs flexible thin-film — affects skin layup and bonding process

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Unidirectional CFRP spar caps (Toray M55J/cyanate-ester prepreg), ~17.5 m half-span — Custom autoclaved spar, similar to port wing but asymmetric.	make	Autoclave Curing · Toray M55J/cyanate-ester prepreg	£17,500
CFRP/Nomex honeycomb sandwich skin panels, autoclave-cured — Large custom sandwich skins with UV-stable outer ply.	make	Autoclave Curing · Carbon fibre/epoxy, Nomex honeycomb	£8,000
Wing-laminated thin-film/IBC mono-Si solar cell array (~25-28% eff., ~12 m²) — Flexible high-efficiency solar array, ~£8000/m² for 12 m².	make	Manual/Assembly · Other	£96,000
Outboard composite motor nacelle with titanium hard-points — Custom nacelle with bonded titanium inserts.	buy	—	£2,800
Twin CFRP control surface set: aileron and outboard spoileron, Kevlar hinged — Custom composite control surfaces with flexure hinges.	make	Autoclave Curing / Assembly · Carbon fibre/epoxy, Kevlar	£4,500
Volz DA-22 stratospheric-rated brushless servo actuators (×2) — Identical to port wing servos, £3000 each.	buy	—	£6,000
Pre-preg foam-core ribs (Rohacell 71IG-F) at ~400 mm pitch — Custom CNC-cut foam ribs with ply drop-offs.	make	CNC Machining / Lamination · Foam	£2,200
Wing root titanium fitting (Ti-6Al-4V) with shear pins — Custom machined interface fitting for structural carry-through.	buy	—	£2,800

Labour	£12,000 — Skilled composite fab & assembly, ~300 hrs at £40/hr.
Per-unit total	**£151,800**

ASSUMPTIONS

- Solar cell cost per m² is lower than GaAs.
- Ribs are CNC-cut foam, not moulded.
- Labour similar to port wing due to symmetry.

Similar to port wing but with lower-cost solar cells and additional ribs.

3.3 · Fuselage & Empennage Structure

Module id: fuselage_empennage — 18.00 kg · 20 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Composite monocoque body and tail surfaces providing the structural backbone, payload bay and stability surfaces.

WHY IT MATTERS

This module is the single load path between the two 17.5 m wing semi-spans and carries every other subsystem (power, propulsion, payload, avionics, gear). Any structural failure is unrecoverable, and any mass overrun cascades into reduced endurance because the HAPS power budget is critically tight. It also defines CG, antenna line-of-sight and bay volumes, so it constrains every downstream integration decision.

DESCRIPTION

The fuselage and empennage form the structural backbone of the Wren Flyer: a slender CFRP monocoque pod approximately 2.5 m long housing the avionics, payload, propulsion bay and hydrogen subsystem, terminated aft by a conventional CFRP empennage with hinged elevator and rudder. Construction is autoclave-cured aerospace prepreg (M55J or T800 fibre in toughened epoxy or cyanate-ester resin) using sandwich panels with Nomex honeycomb or Rohacell foam cores, female-moulded for surface accuracy. A central wing carry-through bulkhead with titanium fittings reacts the bending and shear loads from the 35 m wings, while internal composite frames define the avionics, payload and propulsion bays with discrete bonded inserts and through-bolted hard-points. The outer laminate incorporates a UV-stable surface ply and an embedded conductive mesh for ESD/lightning protection, and is qualified to 70 °C / +60 °C thermal cycling per DO-160G high-altitude categories.

KEY PARTS (6)

- CFRP monocoque fuselage shell, ~2500 mm length, sandwich construction with M55J/cyanate-ester prepreg skins (0.4 mm) over 6 mm Nomex honeycomb core, autoclave-cured in female mould
- Wing-root carry-through bulkhead, machined Ti-6Al-4V fittings bonded/bolted to CFRP frame, transferring ±35 m span bending moment between port and starboard spars
- Conventional empennage: CFRP horizontal stabiliser (~3500 mm span) and vertical fin (~1500 mm height) with hinged elevator and rudder, foam-cored CFRP sandwich skins
- Composite payload bay frame with 25 kg-rated hard-points, MIL-STD-1760-style bay floor with quick-release latches and EMI gasket seal
- Avionics bay with vibration-isolated equipment shelf (wire-rope isolators) and skin penetrations for SATCOM, 1090 ES and GNSS antennas using sealed coaxial bulkhead feedthroughs
- UV-stable outer ply (Tedlar or fluoropolymer film) and conductive lightning-strike mesh embedded in outer laminate

FAILURE MODES (4)

- Wing-root bulkhead delamination or fitting pull-out under gust loading at 20 km, leading to wing separation
- Microcracking of skin laminate from repeated 70 °C to +60 °C thermal cycling, causing moisture ingress and stiffness loss
- UV/ozone degradation of outer ply over multi-week missions, reducing surface integrity and solar array bond line
- Tail boom torsional buckling from rudder loads during STOL crosswind takeoff

OPEN QUESTIONS (3)

- Final fuselage cross-section and length once H2 tank geometry and fuel cell stack envelope are frozen
- Empennage configuration (conventional vs T-tail vs V-tail) — drives tail-boom stiffness and servo routing

- Whether hydrogen tank bay requires an inerted/vented compartment with composite liner compatible with H2 permeation

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
CFRP monocoque fuselage shell (M55J/cyanate-ester, Nomex honeycomb core, ~2500 mm) — Autoclaved sandwich structure, female mould, high-cost materials.	make	Autoclave Curing · M55J/cyanate-ester prepreg, Nomex honeycomb	£12,000
Wing-root carry-through bulkhead machined Ti-6Al-4V fittings — Large complex Ti machined part for major bending load.	buy	—	£8,500
Empennage: CFRP horizontal stabiliser (~3500 mm span) and vertical fin (~1500 mm) — Autoclaved CFRP sandwich tails with hinged surfaces.	make	Autoclave Curing · Carbon fibre/epoxy, Nomex honeycomb	£7,500
Hinged elevator and rudder (foam-cored CFRP sandwich) — Custom control surfaces for empennage.	make	Autoclave Curing / Assembly · Foam	£2,500
Composite payload bay frame with 25 kg-rated hard-points — Custom structural frame with MIL-STD-1760-style interfaces.	make	Composites · Carbon fibre/epoxy	£3,500
MIL-STD-1760-style bay floor with quick-release latches and EMI gasket — Standardised payload interface hardware, COTS latches and seals.	buy	—	£1,800
Avionics bay vibration-isolated equipment shelf (wire-rope isolators) — Fabricated shelf with purchased wire-rope isolators.	make	Sheet Metal / Assembly · Aluminum 6061	£1,200
Skin penetrations for SATCOM, 1090 ES, GNSS antennas with sealed coax bulkhead feedthroughs (×3) — Sealed RF connectors, COTS feedthroughs.	make	Manual/Assembly · Other	£900
UV-stable outer ply (Tedlar or fluoropolymer film) and conductive lightning-strike mesh — Added during autoclave cure of outer skin.	make	Autoclave Curing · Tedlar film, copper mesh	£800

Labour £10,000 — Skilled composite & metal assembly, ~250 hrs at £40/hr.

Per-unit total **£48,700**

ASSUMPTIONS

- Ti bulkhead is large and costly to machine.
- Avionics shelf is simple fabrication.
- Latches and feedthroughs are standard catalogue items.

Monocoque fuselage with empennage and payload interfaces, high composite content.

3.4 · Hybrid Solar-Hydrogen Power System

Module id: hybrid_power_system — 35.00 kg · 28 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Generates, stores and distributes electrical power via wing solar arrays, hydrogen fuel cell stack, H2 tank and regenerative buffer battery.

WHY IT MATTERS

This module determines the aircraft's defining capability: multi-week persistence at 20 km. Insufficient solar capture, fuel-cell efficiency or H2 storage density directly truncates endurance and forces premature recovery, undermining the HAPS value proposition versus LEO satellites. It is also the dominant mass-and-volume consumer after the airframe, so any inefficiency cascades into reduced payload margin or wing loading penalties.

DESCRIPTION

The hybrid power system is a distributed electrical generation, storage and distribution architecture spanning the wing upper surfaces (solar array) and the central fuselage power bay (fuel cell, H2 tank, battery, PMU). Daytime cruise is powered directly by the laminated thin-film solar array via per-wing MPPT controllers, with surplus energy diverted to electrolyser-free regenerative charging of the Li-ion buffer battery. At night and during cloud cover, the PEM fuel cell consumes stored gaseous hydrogen from a Type IV COPV to sustain cruise power, with the battery handling transient loads and providing climb/peak boost. The PMU bus-couples all three sources via redundant ORing diodes and DC-DC converters, producing isolated regulated rails for propulsion (HV), avionics (28 V) and payload (12/28 V).

Key material choices reflect the mass-critical, sub-zero stratospheric environment: ETFE-encapsulated back-contact silicon (or III-V) cells for high specific power and UV/thermal-cycle survival, carbon-fibre/HDPE COPVs for ~6 wt% H2 storage efficiency, silicon-anode Li-ion for energy density >400 Wh/kg, and graphite-composite PEM stacks. All H2-wetted components are 316L stainless or PEEK to mitigate hydrogen embrittlement, and the fuel cell and tank are mounted in a vented, sniffer-monitored bay separated from avionics.

KEY PARTS (6)

- Solar array: ~60 m² of SunPower/Maxeon Gen III back-contact monocrystalline Si cells (24% efficiency) or IMM-triple-junction GaAs cells (32% efficiency), laminated under ETFE film onto upper wing skin, ~5–6 kW peak DC at 20 km solar irradiance
- PEM fuel cell stack: Intelligent Energy IE-SOAR 800W-class or Honeywell HAPS-grade stack scaled to ~2 kW continuous, lightweight graphite/composite bipolar plates, air-breathing with stratospheric compressor pre-conditioning
- Hydrogen storage: Type IV carbon-fibre overwrapped pressure vessel (COPV), HDPE liner, ~350–700 bar, ~8–12 kg usable H2 capacity, integrated solenoid isolation valve and pressure regulator (Luxfer or NPROXX aerospace grade)
- Buffer battery: Li-ion pack using Amprius 450 Wh/kg silicon-nanowire cells, ~2–3 kWh capacity, 28 series modules with integrated BMS, dual-redundant cell-balancing
- Power management unit: custom MPPT array (one per wing half, ~3 kW each), DC-DC converters producing isolated 28 V avionics bus, isolated 12 V payload bus and 270 V HV propulsion bus, CAN/RS-485 telemetry, designed to DO-160G high-altitude category
- Hydrogen balance-of-plant: stratospheric air compressor, humidifier, purge valves, water management/venting system, stainless 316L H2 tubing and Swagelok VCR fittings

FAILURE MODES (4)

- Hydrogen leak from COPV, regulator or tubing fittings, leading to loss of night-time power and potential fuselage bay flammability hazard
- Fuel-cell membrane drying or flooding at stratospheric humidity/temperature extremes, causing voltage collapse and forced descent
- Solar array delamination or micro-cracking from UV exposure and 70 °C to +60 °C thermal cycling on the wing upper skin
- Buffer-battery thermal runaway or capacity fade from deep cycling combined with low-temperature charging, propagating through the pack

OPEN QUESTIONS (3)

- Required total energy budget per 24 h cycle (depends on unpublished propulsion power and payload duty cycle) — drives H2 mass and battery sizing
- Choice between 350 bar and 700 bar gaseous H2 vs cryo-compressed storage, trading tank mass against volume and refuelling complexity
- Whether regenerative electrolysis (closed H2 loop) is in scope or H2 is single-shot consumable per sortie

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Solar cells, laminated on wing skin (60 m²) — Custom lamination of high-efficiency cells onto CFRP wing skin.	make	Lamination · GaAs / Monocrystalline Si cells	£120,000
PEM fuel cell stack (2 kW) — Specialised aerospace fuel cell, limited suppliers, high cost.	buy	—	£28,000
Hydrogen storage COPV, 700 bar, 10 kg H2 — Aerospace grade tank with valve assembly, buy from Luxfer.	buy	—	£15,000
Buffer battery pack, 2.5 kWh, 450 Wh/kg — Amprius cells with custom BMS, buy integrated pack.	buy	—	£12,000
Power management unit (MPPT + converters) — Custom high-altitude rated electronics, DO-160G design.	make	PCB Assembly · FR4 / Aerospace components	£18,000
Hydrogen balance-of-plant (compressor, valves, tubing) — Specialist H2 components, mostly buy (Swagelok, etc.).	buy	—	£22,000

Labour £800 — 20 hours assembly and integration of subsystems.

Per-unit total **£196,800**

ASSUMPTIONS

- Prototype quantities, aerospace grade components.
- Lamination cost includes wing integration preparation.

All parts specified are high-cost aerospace items, with core electronics custom built.

3.5 · Electric Propulsion System

Module id: propulsion_system

8.50 kg · 16 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Wing-mounted brushless motors, controllers and propellers providing thrust for cruise and climb to 20 km.

WHY IT MATTERS

Propulsion sets the achievable cruise altitude, climb time through the jet stream, and night-time hydrogen burn rate. Failure of either motor at 20 km is unrecoverable in cruise (asymmetric thrust on a 35 m span is uncontrollable), so reliability and efficiency directly determine mission viability and the entire energy budget of the hybrid power system.

DESCRIPTION

The propulsion system comprises two wing-mounted electric propulsion units (EPUs), each integrating a high-efficiency brushless permanent-magnet synchronous motor, a high-voltage motor controller, and a large-diameter, low-Reynolds-number variable-pitch propeller optimised for stratospheric cruise where air density is approximately 7% of sea level. Each EPU is housed in a CFRP nacelle bonded to the outboard wing spar via titanium hard-points, with the motor and ESC conduction-cooled to the nacelle skin since convective cooling is negligible at 20 km. Motors are wound for high-altitude operation with derated current and corona-resistant insulation systems; ESCs are conformally coated and potted to suppress partial discharge in low-pressure air.

KEY PARTS (6)

- Brushless outrunner PMSM motors, ~2 kW continuous rated, high-altitude wound (e.g. Launch Point Technologies HALE-class or custom Plettenberg variant), liquid/conduction cooled, qty 2 (one per wing nacelle)
- High-voltage ESCs rated 150 V / 30 A continuous with regenerative braking and CAN-bus telemetry (e.g. APD HV Pro 120F or custom potted unit qualified to DO-160G Cat F altitude)
- Carbon-fibre variable-pitch propellers, 2.0–2.4 m diameter, low-Reynolds high-altitude blade profile (Eppler E854/SD7037-class), 2-blade, custom-moulded by Helix or Mejzlik
- Composite motor nacelle with CFRP firewall, integrated to outboard wing spar via titanium hard-points (Ti-6Al-4V)
- Pitch-change actuator: brushless servo with redundant position sensing, qty 1 per propeller
- Shielded HV DC harness (50–150 V) with corona-resistant PTFE insulation rated for 20 km altitude (per DO-160G Section 4)

FAILURE MODES (4)

- Corona discharge / partial-discharge breakdown in HV windings or ESC at low ambient pressure leading to insulation failure
- Bearing seizure due to lubricant outgassing/thickening at -70 °C stratospheric temperatures
- Variable-pitch actuator jam fixing blade at high-pitch, causing motor stall and asymmetric drag
- Propeller blade flutter or delamination from thermal cycling and UV exposure

OPEN QUESTIONS (3)

- Final motor count and placement (twin wing-mounted vs. distributed electric propulsion across multiple smaller units)
- DC bus voltage selection (28 V LV vs. 270 V HV) trading harness mass against corona risk
- Whether propellers are variable-pitch or fixed-pitch with motor RPM control only

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Brushless outrunner PMSM motors (x2) — High-altitude rated, specialised HALE-class buy.	buy	—	£12,000
High-voltage ESCs (x2) — 150V/30A DO-160G certified buy from APD.	buy	—	£6,000
Carbon-fibre variable-pitch propellers (x2) — Custom moulded by specialist suppliers like Helix.	buy	—	£10,000
Composite motor nacelle with firewall (x2) — Custom CFRP moulding and titanium hardpoints fabrication.	buy	—	£6,000
Pitch-change actuators (x2) — Brushless servo with redundant feedback buy.	buy	—	£4,000
Shielded HV DC harness (corona-resistant) — Custom length PTFE insulated harness assembly.	make	Cable Assembly · PTFE / Copper	£1,200

Labour £800 — 20 hours assembly and integration of drivetrain.

Per-unit total **£40,200**

ASSUMPTIONS

- Prototype build, aerospace quality cable assembly.
- Propellers custom made, per unit cost.

Motors and ESCs dominate cost, nacelle and harness made in-house.

3.6 · Avionics & Flight Control Suite

Module id: avionics_flight_control — 6.50 kg · 16 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Autonomous flight controller, IMU/GNSS, 1090 ES transponder, SATCOM uplink and ground-link radios coordinating the airframe.

WHY IT MATTERS

This module is the cognitive and regulatory core of the platform: without certifiable autonomous flight control, EASA SAIL V-VI BVLOS authorisation and Eurocontrol stratospheric coordination are impossible. Single-string failure at 20 km is unrecoverable, so redundancy architecture and DO-178C/DO-254 assurance directly drive both the airworthiness case and the cost target.

DESCRIPTION

The avionics suite is a triple-redundant, fault-tolerant flight control system housed in a thermally insulated, pressure-compensated avionics bay within the forward fuselage. The core is a custom flight control computer running DO-178C DAL-B autonomy software on lockstep MCUs, fed by a tactical-grade fibre-optic or MEMS IMU and a multi-constellation GNSS (including Galileo PRS for European sovereignty). Air data is provided by heated pitot/static probes and AoA/AoS vanes qualified for -70 °C stratospheric operation. A CAN-FD backbone distributes commands to ~16 brushless servo actuators across wings and empennage and exchanges telemetry with propulsion, power and payload subsystems.

KEY PARTS (7)

- Triple-redundant flight control computer (e.g. custom DAL-B board with NXP MPC5777C lockstep MCUs, 3x voting), DO-178C Level B software, ~150 x 100 x 40 mm
- Tactical-grade IMU with multi-constellation GNSS (e.g. Honeywell HG4930 or Safran STIM377H + u-blox F9P, Galileo PRS-capable for sovereign EU operation)
- 1090 ES Mode S transponder (e.g. Sagetech MX12B or equivalent EASA-certified, Class 1, 250 W output) with top/bottom blade antenna diversity
- Iridium Certus 700 SATCOM terminal for BVLOS C2 link plus secondary L-band line-of-sight datalink modem (AES-256 encrypted, sovereign EU keys)
- Air data computer with heated pitot probe and AoA/AoS vanes rated to -70 °C, DO-160G qualified
- High-torque brushless servo actuators (e.g. Volz DA 22-N) for ailerons, elevator, rudder; CAN bus controlled, 16x distributed across airframe
- CAN-FD avionics backbone with redundant dual-bus topology, MIL-DTL-38999 connectors throughout

FAILURE MODES (4)

- GNSS denial or jamming causing navigation drift — mitigated by IMU dead-reckoning and Galileo PRS
- Thermal shutdown of FCC at -70 °C if bay heater/insulation fails
- SATCOM link loss leaving aircraft on autonomous lost-link contingency for extended period
- Servo actuator stall or runaway driving control surface hardover, requiring independent monitor channel

OPEN QUESTIONS (3)

- Final autonomy DAL allocation (B vs C) pending SORA risk assessment outcome
- Choice between Iridium Certus and EU-sovereign IRIS² SATCOM once operational
-

Whether detect-and-avoid (ACAS-Xu / cooperative-only) is required by Eurocontrol for transit through controlled airspace during climb/descent

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Triple-redundant flight control computer — Custom DO-178C board with lockstep MCUs.	make	PCB Assembly · FR4 / Military components	£15,000
Tactical-grade IMU with GNSS — Honeywell HG4930 or similar, buy COTS.	buy	—	£12,000
1090 ES Mode S transponder — EASA-certified transponder, buy from Sagetech.	buy	—	£5,000
Iridium Certus 700 SATCOM terminal — BVLOS link, buy terminal with antenna.	buy	—	£8,000
Air data computer with heated pitot — DO-160G qualified, buy COTS unit.	buy	—	£6,000
High-torque brushless servo actuators (x16) — Volz DA 22-N, CAN bus, buy per unit.	buy	—	£16,000
CAN-FD avionics backbone with connectors — Custom harness with MIL-DTL-38999 connectors.	buy	—	£4,000

Labour	£1,200 — 30 hours assembly and integration of avionics bay.
Per-unit total	**£67,200**

ASSUMPTIONS

- Prototype quantities, all COTS items at list price.
- Custom flight computer includes software development costs.

Most items are high-cost COTS avionics; backbone harness made custom.

3.7 · ISR & Comms Payload Module

Module id: isr_payload

25.00 kg · 32 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

25 kg modular payload bay housing the EO turret, SIGINT antenna array and secure 5G relay.

WHY IT MATTERS

The payload is the entire revenue/mission justification for the platform — every other subsystem exists to keep these 25 kg aloft at 20 km for weeks. Modularity allows the same airframe to serve multiple customers (defence ISR, civil 5G backhaul, border surveillance) without requalifying the airframe, directly underpinning the <£2.5M unit economics through reuse and customer-funded payload swaps.

DESCRIPTION

The ISR & Comms Payload Module is a 25 kg modular bay occupying the lower-mid fuselage, designed as a swappable LRU with a single composite tray that bolts to four hard-points on the fuselage longerons. It hosts three principal payloads: a gimballed EO/MWIR turret protruding through a heated optical dome on the underside, a distributed SIGINT antenna farm with a centralised RFSoC-based digital receiver, and a downward-facing 5G NR relay panel array. A radiation-tolerant payload management computer aggregates sensor data, applies onboard processing (object detection, geolocation, signal classification), and streams mission product via a Ku-band SATCOM modem while exchanging command/status with the flight controller over redundant CAN and Gigabit Ethernet.

KEY PARTS (6)

- Gimballed EO/IR turret, 4-axis stabilised, ~150 mm diameter, MWIR + visible HD, ~6-8 kg (e.g. Hensoldt Argos-II HD class or Safran Euroflir 410 derivative)
- SIGINT wideband digital receiver, 0.5-18 GHz coverage, conformal wing/fuselage spiral and blade antennas, software-defined backend (e.g. Pentek Quartz RFSoC-based) ~5 kg
- Secure 5G NR relay payload, 3.4-3.8 GHz n78 band, ~10 W TX, AESA or sectored patch array on lower fuselage, integrated AMF/UPF lite (~6 kg)
- Payload management computer, radiation-tolerant SBC (e.g. Xiphos Q8J or Aitech S-A1760 Venus), ARINC-825/CAN + Gigabit Ethernet to flight controller
- Ku-band SATCOM mission data downlink modem, e50 Mbps, AES-256 + Type-1-equivalent crypto module (NATO SECRET capable)
- Composite payload tray with thermally isolated mounts, vibration-isolating wire-rope isolators, MIL-STD-1760 power/data interface to bay backplane

FAILURE MODES (4)

- EO turret gimbal bearing seizure or lubricant outgassing at -70 °C stratospheric soak, causing loss of pointing stability
- Optical dome icing or contamination during ascent through tropopause, blinding EO/IR sensors
- Thermal runaway of dense RF power amplifiers in low-density air where convective cooling is negligible, requiring conductive sink to fuselage skin
- EMI/EMC coupling between wideband SIGINT receiver and on-board 5G transmitter, desensitising SIGINT below noise floor

OPEN QUESTIONS (4)

- Final SIGINT antenna placement and conformal integration with composite skin without compromising structural laminate or stealth signature
- Exportability and ITAR/UK-MOD release status of EO turret options for European-sovereign supply chain
- Required mission data rate ceiling — drives Ku vs Ka-band SATCOM selection and antenna aperture sizing
- Whether the 5G relay must support full gNB functionality or only fronthaul/backhaul, affecting payload computer sizing and power draw

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
Gimballed EO/IR turret — Hensoldt Argos-II HD class, COTS military sensor	buy	—	£450,000
SIGINT digital receiver system (antenna + backend) — Pentek RFSoC + antennas, COTS SIGINT suite	buy	—	£120,000
Secure 5G NR relay payload — AESA sectored array + integrated AMF/UPF lite	buy	—	£85,000
Payload management computer (SBC) — Xiphos Q8J rad-tolerant SBC, COTS military computer	buy	—	£45,000
Ku-band SATCOM modem with crypto module — High-speed mil-grade modem, Type-1 capable COTS	buy	—	£65,000
Composite payload tray with isolators and backplane — CNC machined from CFRP, bonded inserts, wire-rope mounts	buy	—	£3,200
MIL-STD-1760 power/data interface connectors — COTS MIL-STD-1760 connector set, RS Components	buy	—	£1,800
Cables and harness — COTS shielded cables and connectors	buy	—	£1,500

Labour £1,200 — 30 hrs assembly and integration at £40/hr

Per-unit total **£772,700**

ASSUMPTIONS

- COTS sensor and avionics prices from similar programmes
- Tray machining cost based on size and CFRP complexity
- Integration time typical for sensor payload

High-value COTS sensors dominate cost; custom tray is minor make part.

3.8 · STOL Landing Gear

Module id: landing_gear

4.50 kg · 16 wk lead · Specialist judgement

No render generated yet for this module.

PURPOSE

Lightweight retractable gear enabling 200 m runway takeoff and gentle recovery landing.

WHY IT MATTERS

The 200 m STOL constraint is a hard requirement of the platform and drives gear stiffness, tyre selection and brake authority; failure to retract cleanly imposes a parasitic drag penalty that can collapse the night-time energy budget and end the mission, while a landing collapse on recovery destroys a >£2M asset.

DESCRIPTION

The STOL landing gear is a lightweight retractable tricycle configuration optimised for low rolling resistance on a 200 m grass or paved runway and minimal stowed drag during multi-week stratospheric cruise. Two main legs are cantilever CFRP leaf-spring struts mounted to hard-points on the fuselage lower longerons just aft of the CG; a steerable nose leg mounts forward of the payload bay. All three units retract rearward into shallow fairings flush with the fuselage underside via electric linear actuators, eliminating the mass and freezing risk of a hydraulic system at -70 °C. Pneumatic tyres are sized for the 200 m takeoff roll with margin for crosswind and unprepared-strip recovery, and electrically-actuated micro disc brakes on the main wheels provide differential braking for ground steering and short rollout.

KEY PARTS (7)

- Main gear legs: CFRP cantilever leaf-spring struts, ~600 mm length, tuned for 1.2 m/s sink rate energy absorption at ~120 kg MTOM, bonded titanium root fittings
- Main wheels (2x): Aerolite-class 130 mm diameter pneumatic tyres on machined 7075-T6 aluminium hubs with sealed stainless ball bearings, ~150 g per wheel assembly
- Nose wheel assembly: 100 mm diameter castoring/steerable wheel with carbon fork and ±30° steering range driven by Volz DA-22 servo (or equivalent COTS UAV servo)
- Retraction actuators (2x main, 1x nose): Volz DA-26-HT brushless linear actuators, 28 V DC, position-feedback, qualified to -55 °C operation
- Disc brakes (main wheels): Tost or Beringer micro hydraulic calipers with carbon-ceramic discs, electrically actuated via miniature electrohydraulic power pack
- Gear bay doors: 0.5 mm CFRP skin panels with Kevlar hinges, spring-loaded closure, sequenced via microswitches
- Control electronics: Custom landing gear control unit (LGCU) with CAN bus interface to flight controller, 28 V DC input, integrated brake modulation and weight-on-wheels sensing

FAILURE MODES (4)

- Actuator stall at low temperature preventing retraction or extension after stratospheric cold-soak
- Main leg root delamination from repeated landing fatigue cycles at high aspect-ratio wing offload conditions
- Brake seizure or asymmetric braking causing runway excursion during the short 200 m rollout
- Tyre under-inflation or burst on recovery touchdown after multi-week pressure decay at altitude

OPEN QUESTIONS (3)

- Required sink-rate qualification envelope for stratospheric descent recovery (currently estimated, not specified by Wren)
- Whether nose gear must be steerable or free-castoring given differential-thrust ground steering capability

- Final MTOM driving tyre and strut sizing — published only as 25 kg payload, total airframe mass proprietary

COST BREAKDOWN · MEDIUM CONFIDENCE

PART	TYPE	PROCESS / MATERIAL	COST
CFRP main gear legs (2) with titanium fittings — Autoclave-cured CFRP leaf springs, bonded Ti root inserts, complex layup	buy	—	£4,800
Main wheels (2) with 130mm tyres and hubs — Aerolite-class assemblies, COTS UAV wheels	buy	—	£1,200
Nose wheel assembly (100mm, carbon fork) — Machined carbon fork, bonded wheel, custom steerable	make	CNC Machining · Carbon Fiber	£900
Volz DA-22 steer servo — COTS heavy-lift UAV servo, RS Components	buy	—	£1,200
Volz DA-26-HT actuators (3) — COTS brushless linear actuators, 28V, mil-qualified	buy	—	£3,600
Disc brake sets (2) with calipers and hydraulic pack — Tost micro calipers + carbon-ceramic discs, COTS	buy	—	£2,400
Gear bay doors (CFRP) with Kevlar hinges — Thin CFRP skins, bonded Kevlar living hinges	buy	—	£600
Microswitches and sequencing wiring — COTS microswitches, RS Components	buy	—	£200
Landing gear control unit (LGCU) — COTS UAV LGCU with CAN bus, RS Components	buy	—	£1,800
Fasteners and hardware — COTS bolts, nuts, washers, RS Components	buy	—	£150

Labour	£1,600 — 40 hrs assembly, bonding, and rigging at £40/hr
Per-unit total	**£18,450**

ASSUMPTIONS

- Carbon fibre autoclave labour included in make cost
- Actuators and brakes from known UAV suppliers
- LGCU is standard COTS unit

Mix of custom composite structures and COTS actuation/braking components.

4. BOM master (86 rows)

BOM derived from the module decomposition's keyParts, expanded into typed part rows. Part records live in the `parts` table and are joined back to modules via source_module_id.

PART #	NAME	MODULE	TYPE	PROCESS / MATERIAL	MASS	COST
AV-001	Triple-Redundant Flight Control Computer — Triple-redundant flight control computer with voting logic for HALE UAV autonomous operation. Houses three independent processing lanes with cross-channel data link for safety-critical flight control.	Avionics & Flight Control Suite	make	other · 6061 Aluminium · ±±0.2mm	1.80kg	£28,500
AV-002-PUR	Tactical-Grade IMU/GNSS Unit — Tactical-grade fiber-optic gyro IMU with integrated multi-constellation GNSS providing high-accuracy navigation reference. Outputs attitude, position and velocity at 200 Hz to the FCC.	Avionics & Flight Control Suite	buy	—	0.70kg	£32,000
AV-003-PUR	1090 ES Mode S Transponder — 1090 MHz Extended Squitter Mode S transponder with ADS-B Out for civil airspace integration. TSO-C166b compliant for HALE operations.	Avionics & Flight Control Suite	buy	—	0.55kg	£4,800
AV-004-PUR	Iridium Certus 700 SATCOM Terminal — Iridium Certus 700 kbps SATCOM terminal for beyond-line-of-sight command, control and ISR data relay. Provides global L-band coverage for HALE missions.	Avionics & Flight Control Suite	buy	—	1.40kg	£9,500
AV-005-PUR	L-band LOS Datalink Modem — L-band line-of-sight datalink modem providing high-bandwidth secure C2 and ISR downlink to ground station. AES-256 encrypted with frequency hopping.	Avionics & Flight Control Suite	buy	—	0.90kg	£12,500
AV-006-PUR	Air Data Computer — Air data computer measuring static pressure, dynamic pressure, total air temperature and angle of attack/sideslip for stratospheric flight envelope. Provides calibrated airspeed and altitude to FCC.	Avionics & Flight Control Suite	buy	—	0.45kg	£7,800
AV-007	CAN-FD Avionics Backbone Harness — CAN-FD avionics backbone harness interconnecting FCC, IMU, ADC, transponder, datalinks and propulsion ECUs. Shielded twisted pair with MIL-DTL-38999 backshells and split termination.	Avionics & Flight Control Suite	make	other · PTFE-insulated copper · ±±10mm length	1.20kg	£850

AV-008-PUR	MIL-DTL-38999 Connector — MIL-DTL-38999 Series III circular connector for environmentally sealed avionics interconnects. Threaded triple-start coupling with EMI shielding for harsh stratospheric operation.	Avionics & Flight Control Suite	buy	—	0.08kg	£95
AV-ASY	Avionics & Flight Control Suite Assembly — Top-level avionics and flight control suite assembly integrating flight computer, IMU, GNSS-RTK, transponder, SATCOM and radios into a single bay-mounted module.	Avionics & Flight Control Suite	make	other · Aluminium Chassis with Mixed Electronics · ±±0.5mm assembly	3.20kg	£4,200
FE-001	CFRP Monocoque Fuselage Shell — CFRP monocoque fuselage shell forming the primary structural body of the HALE UAV. Co-cured sandwich construction with Nomex honeycomb core providing torsional stiffness and mounting surfaces for wing carry-through, payload bay and empennage.	Fuselage & Empennage Structure	make	other · Carbon Fibre Reinforced Polymer · ±±1.0mm	28.50kg	£18,500
FE-002	Wing-Root Carry-Through Bulkhead — Machined wing-root carry-through bulkhead transferring wing bending and shear loads across the fuselage. Acts as primary attachment frame for both wing assemblies and integrates with fuselage skin via bonded and bolted joints.	Fuselage & Empennage Structure	make	machining · 7075 Aluminium · ±±0.05mm	4.80kg	£1,850
FE-003	CFRP Horizontal Stabiliser — CFRP horizontal stabiliser providing pitch stability at high altitude. Skin-on-spar construction with foam core ribs and integrated elevator hinge bosses.	Fuselage & Empennage Structure	make	other · Carbon Fibre Reinforced Polymer · ±±0.5mm	3.60kg	£3,200
FE-004	CFRP Vertical Fin — CFRP vertical fin providing yaw stability and supporting the rudder. Integrates SATCOM antenna mounting provisions and tail navigation light.	Fuselage & Empennage Structure	make	other · Carbon Fibre Reinforced Polymer · ±±0.5mm	2.40kg	£2,400
FE-005	Elevator Control Surface — Elevator control surface hinged to the horizontal stabiliser trailing edge providing pitch control authority. Lightweight CFRP skin over foam core with embedded servo horn fittings.	Fuselage & Empennage Structure	make	other · Carbon Fibre Reinforced Polymer · ±±0.5mm	0.85kg	£950
FE-006	Rudder Control Surface — Rudder control surface hinged to the vertical fin trailing edge providing yaw control. CFRP skin/foam core construction matched to elevator for parts commonality.	Fuselage & Empennage Structure	make	other · Carbon Fibre Reinforced Polymer · ±±0.5mm	0.55kg	£780

FE-007	Composite Payload Bay Frame — Composite payload bay frame forming a rigid sub-structure inside the fuselage to house the 25 kg ISR payload module. Provides isolation mounts and load paths into the fuselage shell.	Fuselage & Empennage Structure	make	other · Carbon Fibre Reinforced Polymer · ±±0.3mm	3.20kg	£2,600
FE-008	Payload Bay Floor with Quick-Release Hardpoints — CNC-machined payload bay floor plate with NATO-standard quick-release hardpoint pattern enabling rapid swap of EO/SIGINT/5G payloads. Integrates threaded inserts and locating pins for repeatable installation.	Fuselage & Empennage Structure	make	cnc · 7075 Aluminium · ±±0.05mm	2.10kg	£920
FE-009	Avionics Bay Equipment Shelf — CNC-machined avionics bay equipment shelf providing mounting points for flight controller, radios and power electronics. Features integral cable routing channels and isolator pads.	Fuselage & Empennage Structure	make	cnc · Aluminium · ±±0.1mm	0.85kg	£145
FE-010-PUR	Wire-Rope Vibration Isolator — Stainless wire-rope vibration isolator decoupling avionics shelf from fuselage frame, attenuating broadband vibration from props and structural modes.	Fuselage & Empennage Structure	buy	—	0.06kg	£28
FE-011	Lightning-Strike Conductive Mesh Layer — Lightning-strike protection conductive mesh layer co-cured into the fuselage outer skin to dissipate strike currents and protect composite structure.	Fuselage & Empennage Structure	make	other · Expanded Copper Foil · ±±0.05mm thickness	0.45kg	£220
FE-ASY	Fuselage & Empennage Assembly — Top-level fuselage and empennage assembly comprising the composite monocoque body, tail boom, horizontal/vertical stabilizers and integrated payload bay. Forms the structural backbone of the airframe.	Fuselage & Empennage Structure	make	other · Carbon Fiber Reinforced Polymer · ±±0.5mm	28.00kg	£14,500
HP-001-PUR	Monocrystalline Si Solar Cell Array — High-efficiency monocrystalline silicon solar cell array bonded to upper wing skin, supplying primary daytime power. Cells are interconnected in series-parallel strings with bypass diodes.	Hybrid Solar-Hydrogen Power System	buy	—	6.50kg	£8,500
HP-002-PUR	PEM Fuel Cell Stack — PEM fuel cell stack converting stored hydrogen into electrical power for night-time operation and high-load conditions.	Hybrid Solar-Hydrogen Power System	buy	—	5.20kg	£14,500
HP-003-PUR	Type IV Hydrogen COPV Tank — Type IV carbon-fibre overwrapped pressure vessel (COPV) storing gaseous hydrogen at 350 bar for fuel cell feed.	Hybrid Solar-Hydrogen Power System	buy	—	8.00kg	£4,200

HP-004-PUR	Li-ion Buffer Battery Pack — Lithium-ion buffer battery pack smoothing peak loads, providing dawn/dusk transition energy and emergency reserve. Includes integrated BMS.	Hybrid Solar-Hydrogen Power System	buy	—	4.80kg	£1,850
HP-005	Power Management Unit (PMU) — Central Power Management Unit consolidating solar, fuel cell and battery sources, performing MPPT arbitration, bus regulation and load distribution to the 270V HVDC propulsion bus and 28V avionics bus.	Hybrid Solar-Hydrogen Power System	make	other · 6061 Aluminium · ±±0.2mm	2.40kg	£4,200
HP-006	MPPT Controller (per wing) — Per-wing Maximum Power Point Tracking controller optimizing power extraction from the wing-distributed thin-film solar array across stratospheric temperature swings and varying solar incidence.	Hybrid Solar-Hydrogen Power System	make	other · 6061 Aluminium · ±±0.2mm	0.85kg	£1,450
HP-007	DC-DC Converter Module — Isolated DC-DC converter module stepping the 270V HVDC propulsion bus down to 28V avionics and 12V/5V housekeeping rails with active load sharing and stratospheric thermal derating.	Hybrid Solar-Hydrogen Power System	make	other · 6061 Aluminium · ±±0.15mm	0.65kg	£780
HP-008-PUR	Stratospheric Air Compressor — High-altitude air compressor delivering pressurized cathode air to the PEM fuel cell stack at 20 km where ambient density is ~7% sea level. Brushless, oil-free, dynamically balanced for long endurance.	Hybrid Solar-Hydrogen Power System	buy	—	1.40kg	£8,500
HP-009-PUR	Fuel Cell Humidifier — Membrane humidifier conditioning cathode inlet air to maintain PEM membrane hydration during high-altitude low-humidity operation, recovering product water from the exhaust.	Hybrid Solar-Hydrogen Power System	buy	—	0.45kg	£950
HP-010-PUR	Hydrogen Purge Valve — Solenoid-actuated hydrogen anode purge valve cycling residual nitrogen and product water from the fuel cell anode loop. Hydrogen-rated, low-leakage, fail-closed.	Hybrid Solar-Hydrogen Power System	buy	—	0.18kg	£320
HP-011	Water Management Manifold — CNC-machined manifold routing fuel cell product water to condenser and expulsion ports, integrating drain, vent and sensor bosses for the water management subsystem.	Hybrid Solar-Hydrogen Power System	make	cnc · 6061 Aluminium · ±±0.05mm bores, ±0.1mm features	0.32kg	£240

HP-ASY	Hybrid Solar-Hydrogen Power System Assembly — Top-level assembly of the hybrid solar-hydrogen power system integrating PV array, PEM fuel cell stack, H2 COPV tank, Li-ion buffer pack and power management electronics.	Hybrid Solar-Hydrogen Power System	make	other · Mixed Assembly · ±±1.0mm	42.00kg	£38,500
IS-001-PUR	Gimballed EO/IR Turret — Gyro-stabilised gimballed turret with daylight EO camera, cooled MWIR thermal imager and laser rangefinder/designator for high-altitude long-range surveillance and target tracking.	ISR & Comms Payload Module	buy	—	8.50kg	£95,000
IS-002-PUR	SIGINT Wideband Digital Receiver — Wideband software-defined digital receiver (e.g. 20 MHz–18 GHz) for SIGINT/ELINT collection, providing IF digitisation, channelisation and on-board signal classification.	ISR & Comms Payload Module	buy	—	3.20kg	£42,000
IS-003	Conformal SIGINT Spiral/Blade Antenna — Conformal wideband spiral/blade SIGINT antenna formed from etched copper on flexible substrate bonded to a sheet-metal ground plane, mounted flush to the fuselage skin to cover 0.5–18 GHz.	ISR & Comms Payload Module	make	sheet_metal · Aluminium Alloy ground plane with PTFE/copper radiator · ±±0.15mm	0.45kg	£850
IS-004	Secure 5G NR Relay Payload — Airborne secure 5G NR relay payload providing tactical bent-pipe / regenerative connectivity between ground users and the SATCOM backhaul, including baseband processor, RF front-end and dual antenna feeds.	ISR & Comms Payload Module	make	other · Aluminium Alloy enclosure with PCB stack · ±±0.2mm	2.60kg	£12,500
IS-005-PUR	Radiation-Tolerant Payload Management SBC — Radiation-tolerant single-board computer acting as payload manager, scheduling sensors, performing on-board data triage and mission storage at 20 km stratospheric altitude.	ISR & Comms Payload Module	buy	—	0.65kg	£8,500
IS-006-PUR	Ku-band SATCOM Downlink Modem — Ku-band SATCOM modem providing high-rate beyond-line-of-sight downlink of ISR data, with DVB-S2X waveform support and embedded FEC.	ISR & Comms Payload Module	buy	—	1.80kg	£18,500
IS-007-PUR	Crypto Module (AES-256/Type-1 equivalent) — Tamper-protected inline crypto module providing AES-256 (Type-1 equivalent) bulk encryption of payload data and command links between the payload manager and SATCOM modem.	ISR & Comms Payload Module	buy	—	0.55kg	£14,500

IS-008	Composite Payload Tray — Lightweight modular tray supporting EO turret, SIGINT and 5G relay payloads within the ISR bay. Provides isolated mounting points and cable routing channels.	ISR & Comms Payload Module	make	other · Carbon Fibre Reinforced Polymer · ±±0.3mm	1.80kg	£850
IS-009	Heated EO Optical Dome — Optically transparent dome protecting the EO turret with integrated ITO heating film for anti-icing at 20 km altitude. Maintains optical clarity across visible and NIR bands.	ISR & Comms Payload Module	make	other · Optical Acrylic · ±±0.2mm	0.45kg	£620
IS-ASY	ISR & Comms Payload Module Assembly — Top-level integration assembly for the 25 kg modular ISR & Comms payload bay, combining EO/IR turret, SIGINT receiver/antenna, secure 5G relay, SATCOM modem, crypto and management SBC on a shared chassis with thermal and power interfaces.	ISR & Comms Payload Module	make	other · Mixed (Aluminium/CFRP/PCB) · ±±0.5mm	24.50kg	£185,000
LG-001	CFRP Cantilever Main Gear Leg — Cantilever main gear leg in unidirectional carbon fibre, providing energy-absorbing flex during landing impacts. Bonded mounting block interfaces to fuselage frame.	STOL Landing Gear	make	other · Carbon Fibre Reinforced Polymer · ±±0.5mm	1.60kg	£480
LG-002	Main Wheel Hub — CNC-machined wheel hub carrying the pneumatic main tyre and integrating bearings and brake disc mount. Designed for low rolling resistance and minimum mass.	STOL Landing Gear	make	cnc · 6061 Aluminium · ±±0.05mm	0.22kg	£95
LG-003-PUR	Pneumatic Main Tyre — Pneumatic rubber main tyre sized for 200 m unprepared-strip STOL operations. High-flotation profile with reinforced sidewalls for landing impact loads.	STOL Landing Gear	buy	—	0.60kg	£45
LG-004	Nose Wheel Assembly — Steerable nose wheel sub-assembly including hub, tyre, bearings and steering yoke interface. Provides ground directional control during taxi and takeoff roll.	STOL Landing Gear	make	other · Mixed Assembly · ±±0.2mm	0.85kg	£180
LG-005	Carbon Nose Fork — Carbon-fibre nose fork supporting the nose wheel and providing torsional stiffness for steering inputs. Bonded to the steering pivot bearing housing.	STOL Landing Gear	make	other · Carbon Fibre Reinforced Polymer · ±±0.3mm	0.35kg	£220
LG-006-PUR	Volz DA-26-HT Linear Retraction Actuator — High-temperature linear retraction actuator providing extension and retraction force for the main landing gear leg. Sealed brushless DC actuator suitable for high-altitude operation.	STOL Landing Gear	buy	—	0.85kg	£1,250

LG-007-PUR	Micro Hydraulic Disc Brake Caliper — Compact hydraulic disc brake caliper providing wheel braking on landing rollout. Single-piston floating caliper sized for UAV main gear.	STOL Landing Gear	buy	—	0.18kg	£185
LG-008-PUR	Carbon-Ceramic Brake Disc — Lightweight carbon-ceramic brake rotor offering high thermal capacity and low mass for the main landing gear. Vented design dissipates heat after high-energy stops.	STOL Landing Gear	buy	—	0.12kg	£320
LG-009	CFRP Gear Bay Door Panel — Carbon fibre reinforced polymer gear bay door panel that closes flush with the fuselage skin when the gear is retracted to minimise drag at 20 km cruise.	STOL Landing Gear	make	other · Carbon Fibre Reinforced Polymer · ±±0.3mm	0.35kg	£280
LG-010-PUR	Kevlar Door Hinge — Aramid fibre reinforced hinge attaching the gear bay door to the fuselage frame. Provides flex-life and damage tolerance under repeated retraction cycles.	STOL Landing Gear	buy	—	0.04kg	£38
LG-011	Landing Gear Control Unit (LGCU) — Landing Gear Control Unit managing actuator extension/retraction sequencing, brake modulation, weight-on-wheels sensing and squat-switch logic. Communicates with the flight controller over CAN.	STOL Landing Gear	make	other · Aluminium Enclosure with FR4 PCB · ±±0.2mm enclosure	0.25kg	£420
LG-ASY	STOL Landing Gear Assembly — Complete tricycle STOL landing gear assembly with retractable main and nose gear sized for 200 m runway operations. Includes legs, wheels, tyres, retraction actuators and brakes.	STOL Landing Gear	make	other · Mixed Assembly · ±±0.5mm	14.50kg	£4,800
PR-001-PUR	Brushless PMSM Motor (high-altitude wound) — High-efficiency brushless PMSM motor with high-altitude optimized winding for thin-air operation at 20km. Drives variable-pitch propeller via direct shaft coupling.	Electric Propulsion System	buy	—	2.80kg	£1,850
PR-002-PUR	High-Voltage ESC — High-voltage electronic speed controller managing motor commutation and torque from the HV DC bus. Conformal coated for high-altitude low-pressure operation.	Electric Propulsion System	buy	—	0.45kg	£720
PR-003	CFRP Variable-Pitch Propeller Blade — Carbon fibre reinforced polymer variable-pitch propeller blade optimised for thin-air cruise at 20km. Hollow lay-up with foam core for minimum mass at large diameter.	Electric Propulsion System	make	other · Carbon Fibre / Epoxy Composite · ±±0.3mm profile, ±0.1° pitch	0.35kg	£480

PR-004	Propeller Hub — Machined propeller hub mounting variable-pitch blades onto the motor shaft with internal pitch-change mechanism interface. Houses blade root bearings.	Electric Propulsion System	make	machining · 7075 Aluminium · ±±0.025mm critical bores	0.55kg	£320
PR-005	Pitch-Change Actuator — Linear electromechanical actuator driving collective pitch change of the propeller blades. Position feedback via integrated encoder.	Electric Propulsion System	buy	—	0.18kg	£145
PR-006	CFRP Firewall — Carbon fibre reinforced polymer firewall isolating the motor nacelle from the wing structure. Provides thermal and mechanical separation with motor mounting bosses.	Electric Propulsion System	make	other · Carbon Fibre / Epoxy Composite · ±±0.3mm	0.22kg	£95
PR-007	Shielded HV DC Harness — Shielded high-voltage DC harness delivering bus power from PDB to the ESC. EMI-shielded with high-altitude rated insulation to prevent corona discharge.	Electric Propulsion System	make	other · PTFE-insulated Copper with Tinned Braid Shield · ±±5mm length	0.28kg	£65
PR-ASY	Electric Propulsion System Assembly — Top-level assembly of the electric propulsion system integrating wing-mounted brushless motors, ESCs, propellers, mounts and harnessing for cruise and climb to 20 km.	Electric Propulsion System	make	other · Mixed assembly · ±Assembly ±0.5mm	12.50kg	£18,500
PW-001	Port CFRP Spar Caps — Unidirectional CFRP spar caps forming the upper and lower flanges of the primary wing spar, carrying bending loads from wing root to ~80% span. Co-bonded to the shear web during layup.	Port Wing Assembly	make	other · Carbon Fibre Reinforced Polymer · ±±0.3mm thickness, ±1.0mm length	6.20kg	£3,200
PW-002	Port CFRP Sandwich Skin Panels — CFRP/Nomex sandwich skin panels forming the upper and lower aerodynamic surfaces of the port wing. Provides torsional stiffness and supports the bonded solar laminate.	Port Wing Assembly	make	other · CFRP / Nomex Honeycomb Sandwich · ±±0.4mm skin thickness, ±1.5mm panel profile	8.40kg	£4,800
PW-003	Port Foam-Core Ribs — CNC-machined closed-cell foam-core ribs supporting skin panels and maintaining aerofoil profile between spar stations. Wrapped with a single ply of carbon for shear stability.	Port Wing Assembly	make	cnc · Rohacell Structural Foam · ±±0.2mm profile	2.10kg	£850
PW-004	Port Thin-Film GaAs Solar Laminate — Flexible thin-film GaAs photovoltaic laminate bonded to the upper wing skin to generate electrical power for cruise and battery charging at altitude. Encapsulated in fluoropolymer for UV/ozone resistance.	Port Wing Assembly	buy	—	4.80kg	£12,000

PW-005	Port Outboard Motor Nacelle/Pylon — Composite motor nacelle and pylon mounting the outboard brushless propulsion motor to the wing spar at ~70% semi-span. Houses ESC cooling duct and motor wiring.	Port Wing Assembly	make	other · Carbon Fibre Reinforced Polymer · ±±0.5mm at motor mount face	1.60kg	£950
PW-006	Port Titanium Spar Interface Fitting — Machined titanium fitting bonded and bolted to the wing root spar carrying primary bending and shear loads into the fuselage centre section. Mates with corresponding fuselage lug via cross-pin.	Port Wing Assembly	make	machining · Titanium Alloy · ±±0.05mm on bore and bolt pattern	0.85kg	£720
PW-007	Port Aileron — Port aileron control surface providing roll authority, hinged to the wing trailing edge with composite hinges and driven by an internal servo actuator. CFRP skin over foam core for low mass.	Port Wing Assembly	make	other · CFRP / Foam Sandwich · ±±0.5mm profile, ±0.2mm hinge line	1.40kg	£1,100
PW-008	Port Outboard Flaperon — Outboard combined flap-aileron control surface for the port wing, providing roll authority and high-lift augmentation during low-speed STOL operations. CFRP sandwich construction with hinge fittings and servo horn interface.	Port Wing Assembly	make	other · Carbon Fibre Reinforced Polymer · ±±0.5mm	1.80kg	£2,200
PW-009-PUR	Volz DA-22 Stratospheric Servo Actuator — High-altitude rated brushless servo actuator driving the port flaperon, qualified for stratospheric temperature and pressure conditions. Provides positional feedback via redundant magnetic encoders.	Port Wing Assembly	buy	—	0.19kg	£850
PW-010-PUR	Heated Pitot-Static & AoA Vane Assembly — Heated pitot-static probe with integrated angle-of-attack vane providing airspeed, altitude and AoA data to the flight control system. Internal heater prevents icing at stratospheric altitudes.	Port Wing Assembly	buy	—	0.22kg	£1,450
PW-011	Wing Root Titanium Tension/Shear Pin — Primary wing-to-fuselage attachment pin transferring tension and shear loads at the port wing root. Precision machined from aerospace titanium for high strength-to-weight ratio.	Port Wing Assembly	make	machining · Titanium Alloy · ±±0.02mm h6	0.18kg	£180
PW-012-PUR	Self-Aligning Spherical Bushing — Self-aligning spherical plain bushing accommodating wing flex and misalignment at the root pin interface. PTFE-lined for maintenance-free operation in cold stratospheric environment.	Port Wing Assembly	buy	—	0.04kg	£29

PW-ASY	Port Wing Assembly — Top-level port wing assembly integrating spar, skins, ribs, solar laminate, motor pylon, aileron and interface fitting into a single high-aspect-ratio composite lifting surface. Mates to fuselage centre section via titanium root fittings.	Port Wing Assembly	make	other · Carbon Fibre Reinforced Polymer · ±±2.0mm overall span, ±0.5mm at root interface	38.50kg	£28,500
SW-001	Starboard CFRP Spar Caps — Unidirectional carbon fibre spar caps forming the upper and lower load-carrying flanges of the starboard wing main spar. Carries primary bending loads from lift distribution.	Starboard Wing Assembly	make	other · Carbon Fibre Reinforced Polymer · ±±0.3mm thickness	4.20kg	£5,800
SW-002	Starboard CFRP Sandwich Skin Panels — Sandwich skin panels for the starboard wing upper and lower surfaces, providing aerodynamic shape and torsional stiffness. Carbon facesheets bonded to Nomex honeycomb core.	Starboard Wing Assembly	make	other · CFRP/Nomex Sandwich · ±±0.5mm contour	8.60kg	£12,500
SW-003	Starboard Foam-Core Ribs — CNC-machined foam-core ribs forming the internal structural skeleton of the starboard wing. Provides aerodynamic shape and bonds to spars and skin.	Starboard Wing Assembly	make	cnc · Rohacell Foam · ±±0.2mm	0.45kg	£180
SW-004	Starboard IBC Mono-Si Solar Array Laminate — Pre-laminated IBC monocrystalline silicon solar array bonded to upper wing skin to generate cruise and climb power. Provides ~24% efficiency for high-altitude continuous loiter.	Starboard Wing Assembly	buy	—	3.20kg	£4,500
SW-005	Starboard Outboard Motor Nacelle — Composite nacelle housing the starboard outboard brushless motor, ESC and propeller mount with cooling ducts. Aerodynamically faired into the wing leading edge.	Starboard Wing Assembly	make	other · Carbon Fiber Reinforced Polymer · ±±0.3mm	0.85kg	£420
SW-006	Starboard Titanium Hard-Point Fitting — Machined titanium hard-point fitting transferring outboard motor and payload pylon loads into the wing spar. Designed for high-cycle fatigue at 20 km altitude.	Starboard Wing Assembly	make	machining · Titanium Alloy · ±±0.05mm	0.35kg	£280
SW-007	Starboard Aileron — Composite starboard aileron control surface providing roll authority. Hinged to rear spar with carbon torque tube and servo-driven horn.	Starboard Wing Assembly	make	other · Carbon Fiber Reinforced Polymer · ±±0.3mm	0.60kg	£320
SW-008	Starboard Outboard Spoileron — Composite outboard spoileron providing combined roll and lift-dump function for STOL landing and gust alleviation. Actuated by dedicated electromechanical servo.	Starboard Wing Assembly	make	other · Carbon Fiber Reinforced Polymer · ±±0.3mm	0.40kg	£240

SW-009	Starboard Wing Root Titanium Fitting — Machined titanium wing root fitting interfacing the starboard wing spar to the fuselage center section. Carries primary bending and shear loads through tapered pin joints.	Starboard Wing Assembly	make	machining · Titanium Alloy · ±±0.03mm	0.95kg	£620
SW-ASY	Starboard Wing Assembly — Complete right-hand wing assembly mirroring the port wing, integrating spar caps, sandwich skins, control surfaces, solar array provisions and outboard motor pylon mount. Final assembled and balanced unit.	Starboard Wing Assembly	make	other · CFRP composite assembly · ±±1.0mm overall	28.50kg	£48,000

5. Cost waterfall

Cost estimates grounded against the material-properties library (47 rows, freshest 23 Mar 2026) and the `process_capabilities` table (20 rows, freshest unknown). Costs sit in the project's ai_cost_estimates jsonb, keyed by module id.

UNIT COST (ALL-IN)	CEILING (BRIEF)	HEADROOM
£1,473,750	**£2,500,000**	**£1,026,250**

Per-module roll-up

MODULE	COST	% OF UNIT
Port Wing Assembly	£177,900	12.1%
Starboard Wing Assembly	£151,800	10.3%
Fuselage & Empennage Structure	£48,700	3.3%
Hybrid Solar-Hydrogen Power System	£196,800	13.4%
Electric Propulsion System	£40,200	2.7%
Avionics & Flight Control Suite	£67,200	4.6%
ISR & Comms Payload Module	£772,700	52.4%
STOL Landing Gear	£18,450	1.3%

6. Risks register

Every failure mode and open question declared against each module, in one register. Until a dedicated risks store ships with severity + ownership, these rows are the canonical risk inventory.

Port Wing Assembly

KNOWN FAILURE MODES (4)

- Wing root spar delamination or fatigue crack from repeated thermal cycling and gust loads, leading to in-flight structural failure
- Aero-elastic flutter or divergence at altitude due to insufficient torsional stiffness of the thin sandwich skin
- Solar cell laminate cracking/debonding from CTE mismatch with CFRP skin, degrading power generation
- Control surface servo seizure at -70 °C causing loss of roll authority

OPEN QUESTIONS (3)

- Final aspect ratio and chord distribution (washout/twist) pending aero-structural optimisation
- Spar cap layup thickness and ply count to meet ECSS B-basis allowables at 20 km gust case
- Solar cell string topology and MPPT partitioning between port and starboard halves

Starboard Wing Assembly

KNOWN FAILURE MODES (4)

- Aeroelastic flutter or static divergence due to insufficient torsional stiffness at low stratospheric dynamic pressure
- Solar cell delamination or micro-cracking from thermal cycling (-70 °C to +60 °C) and UV exposure over multi-week missions
- Spar cap fibre compression failure at root under gust loading during low-altitude climb-out
- Servo or wiring harness failure at -70 °C causing control surface runaway or lock

OPEN QUESTIONS (3)

- Final aspect ratio and chord distribution (driving spar sizing and rib spacing)
- Whether outboard motor mount is required on this wing or only on port (single vs twin propulsion configuration not published)
- Solar cell technology selection — rigid IBC vs flexible thin-film — affects skin layup and bonding process

Fuselage & Empennage Structure

KNOWN FAILURE MODES (4)

- Wing-root bulkhead delamination or fitting pull-out under gust loading at 20 km, leading to wing separation
- Microcracking of skin laminate from repeated 70 °C to +60 °C thermal cycling, causing moisture ingress and stiffness loss
- UV/ozone degradation of outer ply over multi-week missions, reducing surface integrity and solar array bond line
- Tail boom torsional buckling from rudder loads during STOL crosswind takeoff

OPEN QUESTIONS (3)

- Final fuselage cross-section and length once H2 tank geometry and fuel cell stack envelope are frozen
- Empennage configuration (conventional vs T-tail vs V-tail) — drives tail-boom stiffness and servo routing
- Whether hydrogen tank bay requires an inerted/vented compartment with composite liner compatible with H2 permeation

Hybrid Solar-Hydrogen Power System

KNOWN FAILURE MODES (4)

- Hydrogen leak from COPV, regulator or tubing fittings, leading to loss of night-time power and potential fuselage bay flammability hazard
- Fuel-cell membrane drying or flooding at stratospheric humidity/temperature extremes, causing voltage collapse and forced descent
- Solar array delamination or micro-cracking from UV exposure and 70 °C to +60 °C thermal cycling on the wing upper skin
- Buffer-battery thermal runaway or capacity fade from deep cycling combined with low-temperature charging, propagating through the pack

OPEN QUESTIONS (3)

- Required total energy budget per 24 h cycle (depends on unpublished propulsion power and payload duty cycle) — drives H2 mass and battery sizing
- Choice between 350 bar and 700 bar gaseous H2 vs cryo-compressed storage, trading tank mass against volume and refuelling complexity
- Whether regenerative electrolysis (closed H2 loop) is in scope or H2 is single-shot consumable per sortie

Electric Propulsion System

KNOWN FAILURE MODES (4)

- Corona discharge / partial-discharge breakdown in HV windings or ESC at low ambient pressure leading to insulation failure
- Bearing seizure due to lubricant outgassing/thickening at -70 °C stratospheric temperatures
- Variable-pitch actuator jam fixing blade at high-pitch, causing motor stall and asymmetric drag
- Propeller blade flutter or delamination from thermal cycling and UV exposure

OPEN QUESTIONS (3)

- Final motor count and placement (twin wing-mounted vs. distributed electric propulsion across multiple smaller units)
- DC bus voltage selection (28 V LV vs. 270 V HV) trading harness mass against corona risk
- Whether propellers are variable-pitch or fixed-pitch with motor RPM control only

Avionics & Flight Control Suite

KNOWN FAILURE MODES (4)

- GNSS denial or jamming causing navigation drift — mitigated by IMU dead-reckoning and Galileo PRS
- Thermal shutdown of FCC at -70 °C if bay heater/insulation fails
- SATCOM link loss leaving aircraft on autonomous lost-link contingency for extended period
- Servo actuator stall or runaway driving control surface hardover, requiring independent monitor channel

OPEN QUESTIONS (3)

- Final autonomy DAL allocation (B vs C) pending SORA risk assessment outcome
- Choice between Iridium Certus and EU-sovereign IRIS^2 SATCOM once operational
- Whether detect-and-avoid (ACAS-Xu / cooperative-only) is required by Eurocontrol for transit through controlled airspace during climb/descent

ISR & Comms Payload Module

KNOWN FAILURE MODES (4)

- EO turret gimbal bearing seizure or lubricant outgassing at -70 °C stratospheric soak, causing loss of pointing stability
- Optical dome icing or contamination during ascent through tropopause, blinding EO/IR sensors
- Thermal runaway of dense RF power amplifiers in low-density air where convective cooling is negligible, requiring conductive sink to fuselage skin
- EMI/EMC coupling between wideband SIGINT receiver and on-board 5G transmitter, desensitising SIGINT below noise floor

OPEN QUESTIONS (4)

- Final SIGINT antenna placement and conformal integration with composite skin without compromising structural laminate or stealth signature
- Exportability and ITAR/UK-MOD release status of EO turret options for European-sovereign supply chain
- Required mission data rate ceiling — drives Ku vs Ka-band SATCOM selection and antenna aperture sizing
- Whether the 5G relay must support full gNB functionality or only fronthaul/backhaul, affecting payload computer sizing and power draw

STOL Landing Gear

KNOWN FAILURE MODES (4)

- Actuator stall at low temperature preventing retraction or extension after stratospheric cold-soak
- Main leg root delamination from repeated landing fatigue cycles at high aspect-ratio wing offload conditions
- Brake seizure or asymmetric braking causing runway excursion during the short 200 m rollout
- Tyre under-inflation or burst on recovery touchdown after multi-week pressure decay at altitude

OPEN QUESTIONS (3)

- Required sink-rate qualification envelope for stratospheric descent recovery (currently estimated, not specified by Wren)
- Whether nose gear must be steerable or free-castoring given differential-thrust ground steering capability
- Final MTOM driving tyre and strut sizing — published only as 25 kg payload, total airframe mass proprietary

7. Supplier shortlist (25)

Shortlist built by scoring each supplier in the directory (651 companies) and marketplace listings (28,315 listings) against each module's declared process + material. A low shortlist count usually means the directory doesn't yet have coverage for the project's niche — not that no match exists globally.

mda.space

MDA Space UK

https://mda.space

Contact: sales@mda.space

HQ not declared · ramp role unassigned · match score 37.1

Matched across 1 module: sensor_payload

SCORE BREAKDOWN

total	37.1
keyword	0
process	0
quality	0.4
industry	5
material	0
semantic	21.7
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Industry: Aerospace
- Regulated-industry certs
- Manufacturer

aeronamic.com

Aeronamic

https://aeronamic.com/

Contact: info@aeronamic.com

HQ not declared · ramp role unassigned · match score 44.1

Matched across 3 modules: left_wing, right_wing, Avionics & Flight Control Suite

SCORE BREAKDOWN

total	44.1
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	23.7
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Semantic match
- Industry: Aerospace
- Regulated-industry certs

catalog.orbitaltransports.com

Propulsion & Thrusters

https://catalog.orbitaltransports.com/cubesat-components/propulsion/

HQ not declared · ramp role unassigned · match score 28.9

Matched across 1 module: Electric Propulsion System

SCORE BREAKDOWN

total	28.9
keyword	5
process	0
quality	0.4
industry	0
material	0
semantic	23.5
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

avioaero.com

Avio Aero

https://www.avioaero.com/

Contact: sales@avioaero.com

HQ not declared · ramp role unassigned · match score 44.4

Matched across 1 module: Avionics & Flight Control Suite

SCORE BREAKDOWN

total	44.4
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	24
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Semantic match
- Industry: Aerospace
- Regulated-industry certs

powercellgroup.com

PowerCell Group

https://powercellgroup.com/

Contact: info@powercellgroup.com

HQ not declared · ramp role unassigned · match score 35.9

Matched across 1 module: Hybrid Solar-Hydrogen Power System

SCORE BREAKDOWN

total	35.9
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	25.5
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Industry: Aerospace
- Products

thebusinesslisting.co.uk

ePropelled Systems

https://www.thebusinesslisting.co.uk/places/coventry/epropelled-systems/

Contact: sales@thebusinesslisting.co.uk

HQ not declared · ramp role unassigned · match score 28.7

Matched across 1 module: Electric Propulsion System

SCORE BREAKDOWN

total	28.7
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	28.3
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Products

sabca.be

Sabca

http://www.sabca.be

Contact: info@sabca.be

HQ not declared · ramp role unassigned · match score 42.3

Matched across 2 modules: Fuselage & Empennage Structure, STOL Landing Gear

SCORE BREAKDOWN

total	42.3
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	21.9
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Industry: Aerospace
- Regulated-industry certs
- Aerospace Components & Systems

energyvoice.com

CPH2

https://www.energyvoice.com/renewables-energy-transition/hydrogen/uk-hydrogen/325252/cph2-electrolyser-uk/

HQ not declared · ramp role unassigned · match score 35.0

Matched across 1 module: Hybrid Solar-Hydrogen Power System

SCORE BREAKDOWN

total	35
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	24.6
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Industry: Aerospace
- Products

intelligent-energy.com

Intelligent Energy Ltd

https://www.intelligent-energy.com

Contact: sales@intelligent-energy.com

HQ not declared · ramp role unassigned · match score 45.9

Matched across 1 module: Hybrid Solar-Hydrogen Power System

SCORE BREAKDOWN

total	45.9
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	25.5
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Semantic match
- Industry: Aerospace
- Regulated-industry certs

enpulsion.com

Enpulsion

https://www.enpulsion.com/

HQ not declared · ramp role unassigned · match score 28.4

Matched across 1 module: Electric Propulsion System

SCORE BREAKDOWN

total	28.4
keyword	5
process	0
quality	0.4
industry	0
material	0
semantic	23
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Manufacturing

cooperantennas.com

Cooper Antennas Ltd

https://www.cooperantennas.com/

Contact: sales@cooperantennas.com

HQ not declared · ramp role unassigned · match score 34.4

Matched across 1 module: comms_satcom

SCORE BREAKDOWN

total	34.4
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	24
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Semantic match
- Regulated-industry certs
- Manufacturing

sccomposites.co.uk

South Coast Composites Ltd

https://sccomposites.co.uk

Contact: sales@sccomposites.co.uk

HQ not declared · ramp role unassigned · match score 40.5

Matched across 1 module: Starboard Wing Assembly

SCORE BREAKDOWN

total	40.5
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	20.1
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Industry: Aerospace
- Regulated-industry certs
- Manufacturer

spiritaerosystems.com

Spirit AeroSystems

https://www.spiritaerosystems.com

HQ not declared · ramp role unassigned · match score 43.8

Matched across 1 module: Fuselage & Empennage Structure

SCORE BREAKDOWN

total	43.8
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	23.4
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Semantic match
- Industry: Aerospace
- Regulated-industry certs

nammo.com

Nammo UK Ltd (Nammo Westcott)

https://www.nammo.com/location/westcott/

Contact: sales@nammo.com

HQ not declared · ramp role unassigned · match score 20.4

Matched across 2 modules: Starboard Wing Assembly, Avionics & Flight Control Suite

SCORE BREAKDOWN

total	20.4
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	0
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Industry: Aerospace
- Regulated-industry certs
- Hardware Company

inmarsat.com

Inmarsat

https://www.inmarsat.com

Contact: sales@inmarsat.com

HQ not declared · ramp role unassigned · match score 35.7

Matched across 1 module: comms_satcom

SCORE BREAKDOWN

total	35.7
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	22.3
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Industry: Aerospace
- Manufacturer

brookhouse-aerospace.com

Brookhouse Aerospace Limited

https://brookhouse-aerospace.com/

Contact: enquiries@brookhouse-aerospace.com

HQ not declared · ramp role unassigned · match score 41.8

Matched across 1 module: Fuselage & Empennage Structure

SCORE BREAKDOWN

total	41.8
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	21.4
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Industry: Aerospace
- Regulated-industry certs
- Aerospace Structures & Components

sentinelphotonics.co.uk

Sentinel Photonics

https://www.sentinelphotonics.co.uk/

Contact: info@sentinelphotonics.co.uk

HQ not declared · ramp role unassigned · match score 30.7

Matched across 1 module: ISR & Comms Payload Module

SCORE BREAKDOWN

total	30.7
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	20.3
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Industry: Automotive
- Products

AeroComponents SA

AeroComponents SA

HQ not declared · ramp role unassigned · match score 24.1

Matched across 1 module: Port Wing Assembly

SCORE BREAKDOWN

total	24.1
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	20.7
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Manufacturer

geaerospace.com

GE Aerospace

https://www.geaerospace.com

Contact: defense.marketing@ge.com

HQ not declared · ramp role unassigned · match score 39.5

Matched across 1 module: STOL Landing Gear

SCORE BREAKDOWN

total	39.5
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	19.1
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Industry: Aerospace
- Regulated-industry certs
- Aerospace Engines and Systems

ftgearing.com

FT Gearing Systems Ltd

https://www.ftgearing.com

Contact: sales@ftgearing.com

HQ not declared · ramp role unassigned · match score 39.1

Matched across 1 module: STOL Landing Gear

SCORE BREAKDOWN

total	39.1
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	18.7
capability	0
specialties	0
certifications	10

WHY THEY SCORED WELL

- Industry: Aerospace
- Regulated-industry certs
- Manufacturer

equipmake.co.uk

Equipmake Ltd

https://equipmake.co.uk

Contact: sales@equipmake.co.uk

HQ not declared · ramp role unassigned · match score 32.6

Matched across 1 module: Electric Propulsion System

SCORE BREAKDOWN

total	32.6
keyword	5
process	0
quality	0.4
industry	0
material	0
semantic	24.2
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Semantic match
- Manufacturer

inmarsat.com

Inmarsat Global Ltd

https://www.inmarsat.com

Contact: sales@inmarsat.com

HQ not declared · ramp role unassigned · match score 34.1

Matched across 1 module: comms_satcom

SCORE BREAKDOWN

total	34.1
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	23.7
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Semantic match
- Industry: Aerospace
- Hardware Company

gmv.com

GMV NSL Ltd

https://www.gmv.com

Contact: sales@gmv.com

HQ not declared · ramp role unassigned · match score 33.4

Matched across 1 module: sensor_payload

SCORE BREAKDOWN

total	33.4
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	20
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Industry: Aerospace
- Manufacturer

mobile-industrial-robots.com

MiR

https://mobile-industrial-robots.com/de

Contact: mail@mir-robots.com

HQ not declared · ramp role unassigned · match score 30.0

Matched across 1 module: ISR & Comms Payload Module

SCORE BREAKDOWN

total	30
keyword	0
process	0
quality	0.4
industry	10
material	0
semantic	19.6
capability	0
specialties	0
certifications	0

WHY THEY SCORED WELL

- Industry: Automotive
- Autonomous Mobile Robots (AMR)

in-space.co.uk

In-Space Missions Ltd

https://in-space.co.uk

Contact: sales@in-space.co.uk

HQ not declared · ramp role unassigned · match score 25.1

Matched across 1 module: ISR & Comms Payload Module

SCORE BREAKDOWN

total	25.1
keyword	0
process	0
quality	0.4
industry	0
material	0
semantic	21.7
capability	0
specialties	0
certifications	3

WHY THEY SCORED WELL

- Hardware Company